

A Novel Network With Parallel Resolution Encoders for the Diagnosis of Corneal Diseases

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Abstract—Objective: To propose a deep-learning network for the diagnosis of two corneal diseases: Fuchs' endothelial dystrophy and keratoconus, based on optical coherence tomography (OCT) images of the cornea. **Methods:** In this paper, we propose a novel network with parallel resolution-specific encoders and composite classification features to directly diagnose Fuchs' endothelial dystrophy and keratoconus using OCT images. Our proposed network consists of a multi-resolution input, multiple parallel encoders, and a composite of convolutional and dense features for classification. The purpose of using parallel resolution-specific encoders is to perform multi-resolution feature fusion. Also, using composite classification features enhances the dense feature learning. We implemented other related networks for comparison with our network and performed k-fold cross-validation on a dataset of 16,721 OCT images. We used saliency maps and sensitivity analysis to visualize our proposed network. **Results:** The proposed network outperformed other networks with an image classification accuracy of 0.91 and a scan classification accuracy of 0.94. The visualizations show that our network learned better features than other networks. **Significance:** The proposed methods can potentially be a step towards the early diagnosis of corneal diseases, which is necessary to prevent their progression, hence, prevent loss of vision.

Index Terms—Optical coherence tomography, deep learning, diagnosis, cornea.

I. INTRODUCTION

THE cornea is the transparent part at the front of the eye. It protects the eye and contributes to its refractive power by two-thirds [1], [2]. The cornea has five layers: epithelium, Bowman's, stroma, Descemet's, and endothelium layers [1], [2]. Many diseases affect the cornea and its layers and their thickness. For example, Fuchs' endothelial dystrophy (FED) causes the cornea to be opaque and thick [3]. Keratoconus (KCN) causes the cornea and the Bowman's layer to become thin, and it can lead to loss of the central vision [4], [5]. In the late stages of corneal diseases, corneal transplantation is the only treatment [6], [7].

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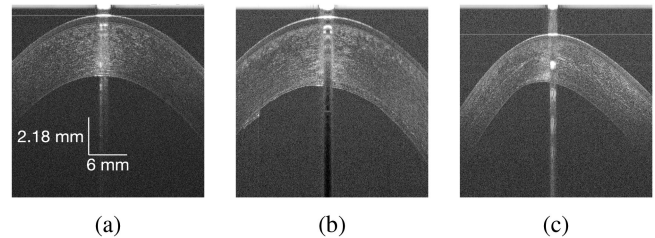


Fig. 1. Examples of optical coherence tomography images for (a) a healthy eye, (b) an eye with Fuchs' endothelial dystrophy, and (c) an eye with keratoconus.

Therefore, the early diagnosis of corneal diseases is necessary to prevent loss of vision.

Optical coherence tomography (OCT) is a non-invasive and non-contact imaging technology, and it can provide high-resolution images of the examined tissue *in vivo* [8]. OCT is widely used in ophthalmology, cardiology, oncology, and dermatology. In ophthalmology, OCT has been used to diagnose several corneal such as keratoconus and Fuchs' endothelial dystrophy [3]–[5] and retinal diseases such as glaucoma and age related macular degeneration [9], [10]. In cardiology, OCT has been used to diagnose coronary artery disease [11], [12]. In oncology, OCT has been used to diagnose cancer and pre-cancerous lesions [13]. In dermatology, OCT has been used to diagnose skin lesions including carcinomas [14]–[16]. Given the advantages of the OCT, OCT images of the cornea can provide high-resolution details of the corneal layers as shown in Fig. 1 (See the supplementary file for more details about the corneal layers). Existing methods diagnose the corneal diseases using maps derived from the OCT scans. These maps include thickness, curvature, elevation, and pachymetry maps. For example, FED is diagnosed with a thickened Descemet's membrane [17]. KCN is diagnosed with a thin cornea, a thin epithelium layer, and a thin Bowman's layer [18]–[20]. However, obtaining these maps requires accurate segmentation of the corneal layers. Also, these maps do not use the OCT image features such as pixel intensity values.

Deep learning has become the state-of-the-art in image classification tasks. Deep networks can learn hierarchical non-linear convolutional features by processing the image through a series of convolutional and pooling layers. Then, fully connected layers are added to learn dense features from the convolutional features. A Softmax layer is used to obtain the classification features based on the dense features [21], [22]. There are many classification networks such as VGG16 [23], Inception [24], ResNet50 [25],

and Xception [26] that have achieved state-of-the-art results on the ImageNet dataset. Deep learning has revolutionized the medical field and has been used in many medical studies to predict several diseases [27]–[30]. For example, deep learning is used to predict Alzheimer's disease in brain magnetic resonance imaging (MRI) [27] and pulmonary nodules in lung computed tomography (CT) [28]. In ophthalmology, several eye disease are detected using deep learning on different imaging modalities [31]–[35]. For example, diabetic retinopathy is detected in retinal fundus images [36]. Age-related macular degeneration is predicted in OCT images [37], [38]. For the diagnosis of corneal diseases, Kamiya *et al.* [39] train a network using six color-coded maps to detect KCN. They use elevation and curvature maps of both anterior and posterior surfaces of the cornea. Also, they use total refractive power, and pachymetry maps. They achieved an accuracy of 99.3% in detecting KCN. Yousefi *et al.* [40] use axial, refractive, elevation, and pachymetry maps of both anterior and posterior surfaces of the cornea. They extract features from these maps using PCA. They cluster the important features to identify healthy eyes from eyes with KCN. They achieved a specificity of 94.1% and a sensitivity of 97.7%. Kovacs *et al.* [41] use a multilayer perceptron with pentacam indices to identify eyes with KCN. They reported area under the receiver operating characteristic (ROC) curve (AUC) of 0.99, a sensitivity of 100%, and a specificity of 95%. However, none of these two approaches use corneal OCT images. To the best of our knowledge, only Treder *et al.* [42] used deep learning with OCT images to detect graft detachment. They use transfer learning with a pre-trained network. They reported an accuracy of 96%. In our study in [43], we fine-tuned pre-trained AlexNet, VGG16, and VGG19 networks using OCT images to diagnose three corneal diseases: dry eye syndrome, FED, and KCN, versus healthy controls (HCN) where we achieved a diagnosis accuracy of 88.75% on the image level and 90% on the scan level.

In this paper, we propose a novel network with parallel resolution-specific encoders and composite classification features to directly diagnose FED and KCN using OCT images. The proposed network consists of a multi-resolution input, parallel and similar encoders, and a composite of convolutional and dense features for classification. The multi-resolution input enlarges the receptive field processed by the network, which makes the network robust. Each encoder consists of the convolutional layers of a pre-trained VGG16 network. Resolution-specific feature maps are obtained by encoding the input at the different resolutions using the parallel encoders. Multi-resolution feature maps are obtained by properly upsampling, concatenating, and reducing, depth-wise, the resolution-specific feature maps. Using parallel resolution-specific encoders helps to achieve multi-resolution feature fusion. Dense features are generated using a fully connected layer. A composite classification features are obtained by concatenating the fused multi-resolution convolutional features with the dense features. Using composite classification features enhances the dense feature learning. For comparison, we constructed simpler networks having some components of our network. We obtained a dataset of OCT scans from patients recruited at the Bascom Palmer Eye Institute (BPEI), and the scans were labeled by corneal specialists from BPEI. We

performed k-fold cross-validation on the dataset. To compare all the networks, we plotted the average of the ROC curves of all the classes. Also, we used the bootstrapping method to estimate the 95% confidence interval (CI) for the AUC. In addition, we compared the architecture, and training and testing times of all considered networks. Finally, we visualized the proposed network using image-specific saliency maps and performed a sensitivity analysis to show how it works. The results show that the proposed network outperformed the other networks and learned better features. The contributions of this paper include:

- Diagnosing two major corneal diseases: Fuchs' endothelial dystrophy and keratoconus directly using OCT images.
- Proposing a novel network with parallel resolution-specific encoders, which helps to achieve multi-resolution feature fusion.
- Using composite classification features, which enhances the dense feature learning.

The paper is organized as follows: Section II describes the proposed methods. Section III describes the data and the experiments. Section IV presents the results. Section V discusses the results and highlights the advantages and limitations of our work. Section VI concludes the paper.

II. PROPOSED METHODS

We define the diagnosis of corneal diseases as a classification problem with multiple classes. Each class is assigned a class label $C \in \{0, 1, 2\}$ where the labels refer to FED, HCN, and KCN, respectively. Each image is labelled with one-hot encoding vector (e.g., keratoconus label is $[0, 0, 1]$). A network is trained when it maps the input image to the prediction probabilities that are close to the one-hot-encoding of the ground truth. The details of the architecture of the proposed network are illustrated in Fig. 2.

A. Data Oversampling

The classes in our dataset had imbalanced number of images, so we expanded the classes with fewer images using oversampling [44]. Oversampling was done by randomly sampling, without replacement, an equal number of images from each class in every batch (i.e., the portion of the dataset used to update the training of the network). We calculated the number of batches B in the training dataset by

$$B = \frac{1}{|C|} \sum_{c=1}^{|C|} n_c \quad (1)$$

where $|C|$ is the number of classes and n_c is the number of images in class c . If each class c has k images in every batch (i.e., batch size = $k |C|$), the probability that a chosen image of class c exists in the batch is given by

$$p = \binom{n_c}{k} / \binom{n_c - 1}{k - 1} = k/n_c \quad (2)$$

Thus, the probability of sampling images from the classes with fewer images (i.e., smaller n_c) is higher, which can be thought of as *implicit* class weights that handles the class imbalance. The

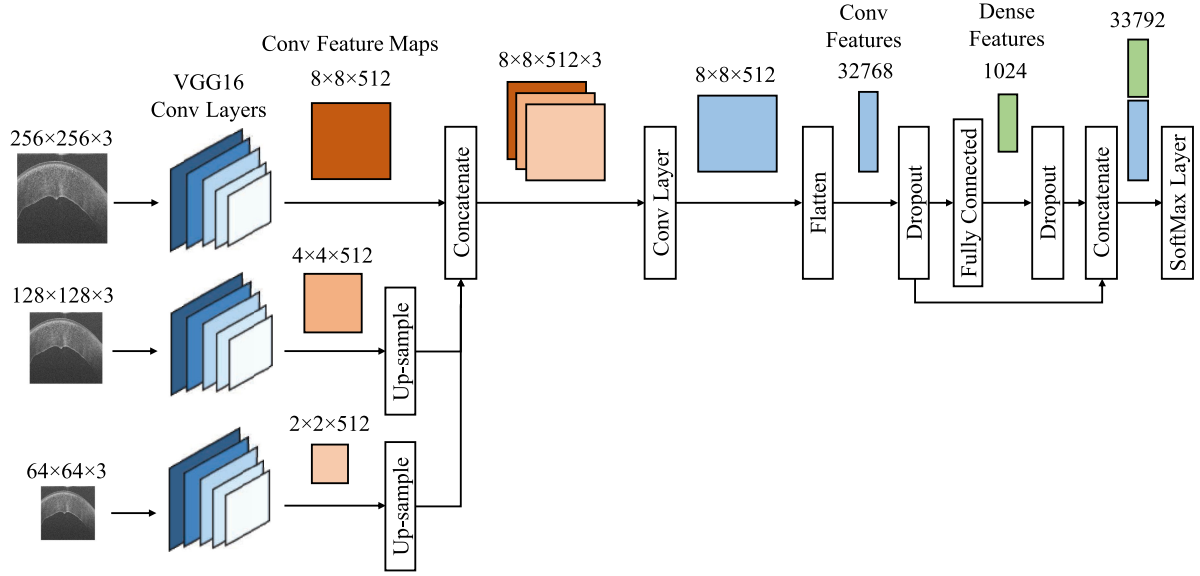


Fig. 2. The proposed multi-scale network where the feature are extracted from each image scale using VGG16 convolutional part, the multi-scale feature maps are concatenated and reduced using a pointwise convolution, dense features are generated using a fully connected layer, and convolutional and dense features are connected to the SoftMax layer.

TABLE I

THE ARCHITECTURE OF THE ENCODER WHICH CONSISTS OF THE CONVOLUTIONAL LAYERS OF THE VGG16 NETWORK [23]

Encoder Architecture	
Layer	Description
Convolutional 1	$\{3 \times 3 \times 64\}$ conv $\times 2$
MaxPooling	2×2
Convolutional 2	$3 \times 3 \times 128$ conv $\times 2$
MaxPooling	2×2
Convolutional 3	$3 \times 3 \times 256$ conv $\times 3$
MaxPooling	2×2
Convolutional 4	$3 \times 3 \times 512$ conv $\times 3$
MaxPooling	2×2
Convolutional 5	$3 \times 3 \times 512$ conv $\times 3$
MaxPooling	2×2

probability that a specific image exists in all batches is given by

$$p^B = \left(\frac{k}{n_c} \right)^B \quad (3)$$

This probability is very low and it approaches zero after training for many epochs, which increases the probability of using all the images during the training.

B. Proposed Network

The proposed network has a multi-resolution input, parallel resolution-specific encoders, and composite classification features. We use decimation layers to generate the lower resolutions of the input image. Using lower resolution images helps to increase the receptive field processed by the network. We use the convolutional layers of a pre-trained VGG16 network [23] as the encoder of our network (See the supplementary file for more details about the selection of the pre-trained network). The architecture of the encoder is summarized in Table I. Resolution-specific feature maps are obtained by encoding the input at the different resolutions using the parallel encoders.

Multi-resolution feature maps are obtained by properly upsampling and concatenating the resolution-specific feature maps. We used nearest neighbor interpolation for the upsampling. The multi-resolution feature maps are reduced, depth-wise, using a point-wise convolutional layer with a linear activation function. Using parallel resolution-specific encoders helps to achieve multi-resolution feature fusion. The multi-resolution feature maps make the network more sensitive to the high-level features, which are present at all resolutions, and less sensitive to noise, which is present at the highest resolution. The fused multi-resolution convolutional feature maps have a size of $8 \times 8 \times 512$. A fully connected layer with 1024 neurons and a ReLU activation function is used to generate dense features of size 1024×1 . Composite classification features are obtained by concatenating the multi-resolution convolutional features and the dense features. Finally, we add a classification layer with three neurons representing the output classes and a Softmax activation function. The architecture of the proposed network is summarized in Table II.

C. Loss Function

We used a categorical cross-entropy loss function to train the parameters of the proposed network which is given by

$$\mathcal{L} = - \sum_c \sum_i^{n_c} g_{c,i} \log(p_{c,i}) \quad (4)$$

where the summations are over $|\mathcal{C}|$ classes and n_c images per class. The predicted categorical label is $p_{c,i} \in P$ and the ground truth label is $g_{c,i} \in G$. The categorical loss is differentiable and its derivative is given by

$$\frac{\partial \mathcal{L}}{\partial p_{c,j}} = - \frac{g_{c,j}}{p_{c,j}} \quad (5)$$

TABLE II

THE ARCHITECTURE OF THE PROPOSED MODEL WHERE ALL LAYERS ARE STACKED SEQUENTIALLY

Main Architecture	
Layer	Output size
Input scale 1	$256 \times 256 \times 3$
Input scale 2	$128 \times 128 \times 3$
Input scale 3	$64 \times 64 \times 3$
VGG sub-network 1	$8 \times 8 \times 512$
VGG sub-network 2	$4 \times 4 \times 512$
VGG sub-network 3	$2 \times 2 \times 512$
Nearest Neighbor Upsampling	$8 \times 8 \times 512$
Concatenation	$8 \times 8 \times 1536$
Convolution	$8 \times 8 \times 512$
Flatten	32768×1
Dropout (0.5)	32768×1
Fully connected	1024×1
Dropout (0.5)	1024×1
Concatenation	33792×1
Softmax	1×3

TABLE III

DETAILED DESCRIPTION OF THE DATASET USED IN THIS STUDY

#	Normal	Fuchs' Dystrophy	Keratoconus	Total
Images	3,677	2,297	10,747	16,721
Patients	67	30	161	258
Eyes	112	48	261	421
Scans	131	70	306	507

follows:

$$\mathcal{S} = \sum_{ch=1}^{|\mathcal{CH}|} W^{ch} \otimes \mathcal{F}_{conv}^{ch} \quad (7)$$

where the summation is over the number of channels $|\mathcal{CH}|$ and $\otimes$ denotes element-wise product. In our proposed network, the convolutional features are connected to the classification layer through the fully connected layer and directly; therefore, the weighted features are calculated using

$$\mathcal{S} = \sum_{ch=1}^{|\mathcal{CH}|} (W_1^{ch} + W_2^{ch}) \otimes \mathcal{F}_{conv}^{ch} \quad (8)$$

where W_1^{ch} is the projected weights from the classification layer for channel ch and is given by (6), and W_2^{ch} is the direct weights of the classification layer for channel ch . Finally, the saliency map $\mathcal{S}$ is resized to the size of the input image and overlaid on it. We also perform a sensitivity analysis [46] of the proposed network output by occluding parts of the input image and re-classifying the image again to reveal which parts are important for classification. A heatmap is constructed using the network output score corresponding to the predicted class. This heatmap weights the pixels of the input image by their importance. To find the best visualizations, we use occluded parts of the input image with different sizes.

III. EXPERIMENTS

D. Visualizations

To visualize the networks and gain insight on how they work, we generate image-specific saliency maps [45]. We also perform a sensitivity analysis [46] for the proposed network. The generation of the saliency maps is summarized in Fig. 3. First, the weights of the fully connected layers, corresponding to the output class, are projected back to the last convolutional feature maps using

$$W = W_{Softmax}^c \cdot (W_{dense} \cdot A) \quad (6)$$

where $W_{Softmax}^c$ denotes the weights of the Softmax classification layer corresponding to the predicted class c , W_{dense} denotes the weights of the fully connected layer, and the vector A is used to account for the output non-linearity where it has ones if the activation of the fully connected layer is greater than zero and zeros otherwise. The final weights W are reordered into the shape of the convolutional feature $\mathcal{F}_{conv}$ and multiplied element-wise by their activation. The saliency map $\mathcal{S}$ is calculated as the point-wise sum of all weighted feature maps as

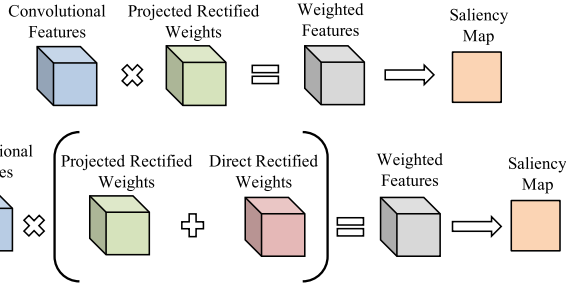


Fig. 3. The generation of the saliency maps for a conventional network (top) and the proposed network (bottom). In the conventional network, the weights of the classification layer are projected back to the convolutional feature maps and multiplied by them. In the proposed network, direct weights are added to the projected weights. Finally, the weighted features are reduced depth-wise.

This ensures that the model can be trained using back-propagation.

A. Data Description

Radial OCT scans of the cornea were obtained from 421 eyes of 258 patients using a high-resolution OCT machine (Envisu R2210, Bioptigen, Buffalo Grove, IL, USA). The patients were recruited at the Bascom Palmer Eye Institute (BPEI). Each OCT scan contained 36 images, and we removed the badly captured images from each scan. For some eyes, there are multiple scans from multiple visits to follow up. OCT scans were diagnosed by corneal specialists from BPEI. The details of the data are summarized in Table III. The obtained OCT images have a size of 1024×1000 pixels, but due to memory considerations, the OCT images were resized to 256×256 pixels. We set the image dimensions to a power of two to facilitate the incorporation of multiple image resolutions into our network.

B. Evaluation

The performance of all the networks was evaluated using accuracy $\mathcal{A}$, sensitivity $\mathcal{SN}$, specificity $\mathcal{SP}$, F1-score $\mathcal{F1}$, and area under ROC curve for the false positive range 0 – 10%.

TABLE IV

THE DESCRIPTION OF THE PROPOSED NETWORK AND OTHER NETWORKS USED FOR COMPARISON

Network	# Input Resolutions	# FCL(s)	$ \mathcal{F}_{dense} $	$\mathcal{F}_{class}$
Conv	1	-	-	$\mathcal{F}_{conv}$
Dense-S	1	1	1024	$\mathcal{F}_{dense}$
Dense-M	1	2	1024	$\mathcal{F}_{dense}$
Dense-L	1	2	2048	$\mathcal{F}_{dense}$
Composite	1	1	1024	$\mathcal{F}_{conv}, \mathcal{F}_{dense}$
Proposed	3	1	1024	$\mathcal{F}_{conv}, \mathcal{F}_{dense}$

FCL denotes fully connected layer, $\mathcal{F}_{class}$ denotes the features used by the classification layer, $\mathcal{F}_{conv}$ denotes the convolutional features, and $\mathcal{F}_{dense}$ denotes the dense features

These measures are defined as follows:

$$\mathcal{A} = \frac{TP + TN}{TP + FP + TN + FN} \quad (9)$$

$$\mathcal{SN} = \frac{TP}{TP + FN} \quad (10)$$

$$\mathcal{SP} = \frac{TN}{TN + FP} \quad (11)$$

$$\mathcal{F1} = \frac{2TP}{2TP + FP + FN} \quad (12)$$

where TP , FP , TN , and FN are the true positive, false positive, true negative, and false negative rates, respectively.

C. Setup

All the networks in this study were implemented using python 3.6.8 and Keras library version 2.2.4 with Tensorflow backend version 1.12.0. The experiments were performed on Windows 10, Home edition 64 bits, running on a PC equipped with a processor Intel(R) Core(TM) i7-8700 K CPU@ 3.70 GHz, a 16 GB RAM, an NVIDIA GeForce GTX 1080 Ti 11 GB. We augmented the OCT images online with a probability of 0.5 using random translations, rotations, and top region erasing where the top region usually contains saturation artifact. We used a batch size of 24 for both training and validation data. To train the networks, the training loss was optimized using stochastic gradient descent (SGD) optimizer [47] with a learning rate of $1E^{-4}$ and a decay rate of $1E^{-6}$. However, with **Dense-L** and **Composite** described in Table IV, we used a learning rate of $1E^{-5}$. We used *early stopping* to stop the training if the training loss did not decrease within five epochs, and we set the maximum number of training epochs to 50 epochs. To evaluate the networks, K -fold cross-validation, with $K = 5$, was performed where $(K - 1)$ folds were used for training, and one fold was used for testing. The performance measures were computed for all testing folds, and the mean $\pm$ standard deviation scores are reported. For the sensitivity analysis of the proposed network, we used different sizes for the occluded part of the input OCT image, and we report the best result along with the used window size.

D. Comparison

For a comprehensive evaluation of our proposed network, we constructed simpler networks that contained one or more

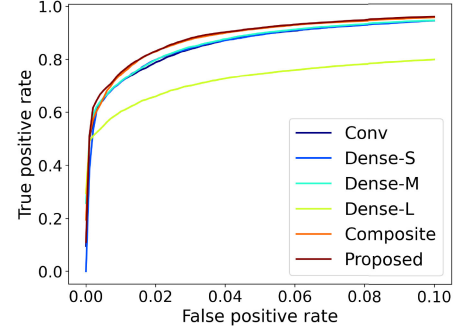


Fig. 4. The averaged receiver operating characteristic (ROC) curve across all folds where the proposed network has the best performance.

components of our network. The descriptions of all the networks are summarized in Table IV. The number of the input image resolutions, the number of fully connected layers, and the features used for the classification are reported for each network. All the networks were trained and evaluated using our dataset. To compare all the networks, we report the average of all the ROC curves and compute their AUCs. To show the performance of the proposed network for each class, we report the class ROC curves and compute their AUCs. Also, we show a visualization of the best-performing networks using saliency maps and perform a sensitivity analysis of the proposed network as described in Section II-D to gain insight how they work.

IV. RESULTS

A. Network Comparison

To show the average performance of each network in classifying all the images, we plotted the average of the ROC curves of all the classes over all folds as shown in Fig. 4. The average $\pm$ standard deviation for the accuracy, sensitivity, specificity, F1-score, and AUC scores are summarized in Table V for all the networks defined in Section III-D. The proposed network outperformed the other networks with scores of 0.91 ± 0.01 , 0.91 ± 0.01 , 0.95 ± 0.0 , 0.91 ± 0.0 , and 0.94 ± 0.01 for the accuracy, sensitivity, specificity, F1-score, and AUC, respectively. **Composite** network, which combines convolutional and dense features, performed approximately (i.e., due to rounding) the same as the proposed network with scores of 0.91 ± 0.01 , 0.91 ± 0.01 , 0.96 ± 0.0 , 0.91 ± 0.0 , and 0.94 ± 0.01 for the accuracy, sensitivity, specificity, F1-score, and AUC, respectively.

B. Class ROC Curves

To show the performance of the proposed network for classifying the images of each class, we plotted the ROC curve of each class as shown in Fig. 5(a). The sensitivity, specificity, F1-score, and AUC scores are summarized in Table VI. The proposed network achieved sensitivity scores of 0.91 ± 0.05 , 0.85 ± 0.07 , and 0.93 ± 0.02 , specificity scores of 0.99 ± 0.0 , 0.94 ± 0.01 , and 0.90 ± 0.05 , F1-scores of 0.94 ± 0.03 , 0.82 ± 0.03 , and 0.94 ± 0.01 , and AUC scores of 0.99 ± 0.01 , 0.86 ± 0.03 , and 0.93 ± 0.02 for FED, HCN, and KCN classes, respectively. The confusion matrix of the proposed network is shown in Fig. 5(b) where the proposed network achieved accuracy values of 0.91,

TABLE V
THE CROSS-VALIDATION RESULTS FOR ALL NETWORKS

Network	Accuracy	Sensitivity	Specificity	F1-score	AUC
Conv	0.88 ± 0.03	0.90 ± 0.01	0.95 ± 0.0	0.90 ± 0.0	0.92 ± 0.01
Dense-S	0.88 ± 0.03	0.90 ± 0.02	0.94 ± 0.0	0.88 ± 0.0	0.92 ± 0.02
Dense-M	0.90 ± 0.02	0.90 ± 0.01	0.94 ± 0.0	0.88 ± 0.0	0.93 ± 0.01
Dense-L	0.84 ± 0.13	0.83 ± 0.14	0.95 ± 0.0	0.91 ± 0.0	0.85 ± 0.17
Composite	0.90 ± 0.02	0.91 ± 0.01	0.96 ± 0.0	0.91 ± 0.0	0.94 ± 0.01
Proposed	0.90 ± 0.03	0.91 ± 0.01	0.95 ± 0.0	0.91 ± 0.0	0.94 ± 0.01

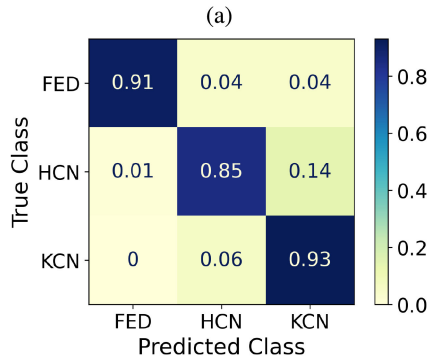
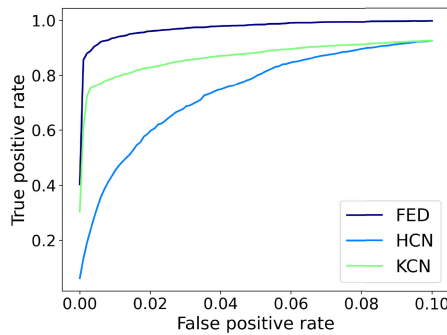


Fig. 5. (a) The image-level receiver operating characteristic curve and (b) the image-level confusion matrix of all classes obtained using the proposed network where FED denotes Fuchs' endothelial dystrophy, HCN denotes healthy controls, and KCN denotes keratoconus.

TABLE VI

THE CROSS-VALIDATION RESULTS OF THE PERFORMANCE SCORES OF THE PROPOSED NETWORK FOR EACH CLASS WHERE THE PERFORMANCE IS MEASURED PER IMAGE

Class	Sensitivity	Specificity	F1-score	AUC
FED	0.91 ± 0.05	0.99 ± 0.0	0.94 ± 0.03	0.99 ± 0.01
HCN	0.85 ± 0.07	0.94 ± 0.01	0.82 ± 0.03	0.86 ± 0.03
KCN	0.93 ± 0.02	0.90 ± 0.05	0.94 ± 0.01	0.93 ± 0.02

0.85, and 0.93 for FED, HCN, and KCN classes, respectively. The proposed network performed well for all classes and the best performance was obtained for the FED class and the lowest performance was obtained for the HCN class. Although, FED had the least number of images in our dataset, it had the best performance due to the presence of clear signs of the disease in the OCT images of our dataset. The HCN had the least performance because the dataset was dominated by the KCN as described in Table III. Also, in early stages of KCN, the OCT images have mild signs of the disease, and they look similar to

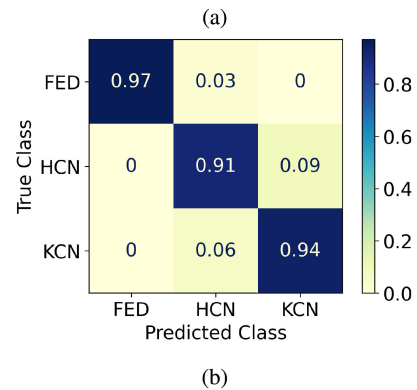
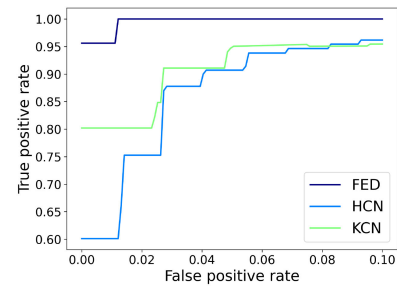


Fig. 6. (a) The scan-level receiver operating characteristic curve and (b) the scan-level confusion matrix of all classes obtained using the proposed network where FED denotes Fuchs' endothelial dystrophy, HCN denotes healthy controls, and KCN denotes keratoconus.

TABLE VII

THE CROSS-VALIDATION RESULTS OF THE PERFORMANCE SCORES OF THE PROPOSED NETWORK FOR EACH CLASS WHERE THE PERFORMANCE IS MEASURED PER SCAN

Class	Sensitivity	Specificity	F1-score	AUC
FED	0.94 ± 0.05	1.0 ± 0.0	0.96 ± 0.03	1.0 ± 0.0
HCN	0.91 ± 0.05	0.95 ± 0.02	0.88 ± 0.02	0.93 ± 0.04
KCN	0.94 ± 0.03	0.94 ± 0.03	0.95 ± 0.01	0.95 ± 0.02

HCN. HCN was misclassified as KCN at a rate of 14% as shown in the confusion matrix because of similarity between HCN and early-stage KCN.

The final diagnosis for the eye was obtained based on the predictions of all the images in the scan. We averaged the prediction probabilities of the network over all the images in the scan, and the predicted class was obtained as the class with the maximum probability. To show the performance of the proposed network for classifying the scans of each class, we plotted the ROC curve of each class as shown in Fig. 6(a). The sensitivity, specificity, F1-score, and AUC scores are summarized in Table VII. The proposed network achieved sensitivity

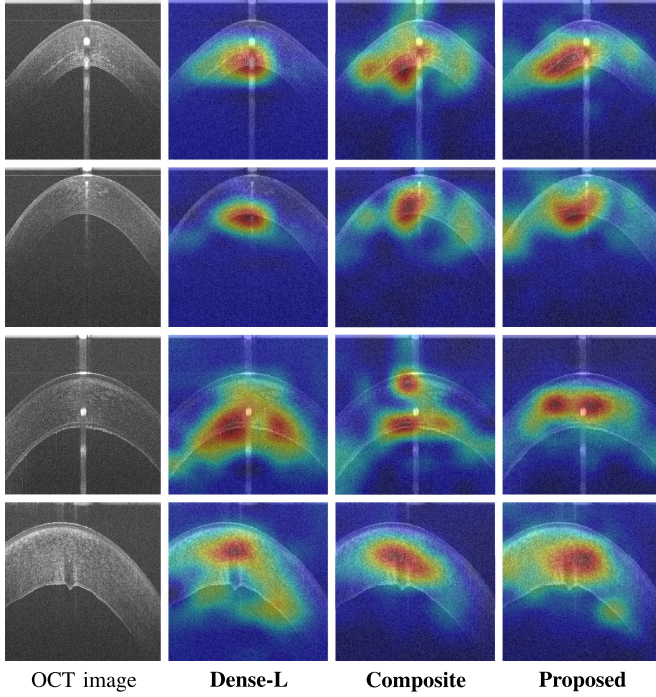


Fig. 7. Visualization examples of the weighted convolutional features overlaid on the optical coherence tomography images for Dense-L, Composite, and Proposed networks where the red and blue colors represent the most and least important regions, respectively.

scores of 0.94 ± 0.05 , 0.91 ± 0.05 , and 0.94 ± 0.03 , specificity scores of 1.0 ± 0.0 , 0.95 ± 0.02 , and 0.94 ± 0.03 , F1-scores of 0.96 ± 0.03 , 0.88 ± 0.02 , and 0.95 ± 0.01 , and AUC scores of 1.0 ± 0.0 , 0.93 ± 0.04 , and 0.95 ± 0.02 for FED, HCN, and KCN classes, respectively. The confusion matrix of the proposed network is shown in Fig. 6(b). The proposed network achieved accuracy values of 0.97, 0.91, and 0.94 for FED, HCN, and KCN classes, respectively. These results ensured that final diagnosis is more accurate in detecting the corneal diseases.

C. Feature Visualization

To gain insight how the networks worked, we visualized their internal activity using saliency maps described in Section II-D and overlaid them on the input OCT images. We visualized the saliency maps for **Proposed** (i.e., the best network), **Composite** (i.e., the second-best), and **Dense-L** (i.e., the third-best) as shown in Fig. 7. We used the jet colormap for visualizations where the red color represents the most important regions, and the blue color represents the least important regions. In the top two OCT images of Fig. 7, the features activated by **Dense-L** had less regions of the anterior surface of the cornea. Contrarily, the features activated by **Composite** and **Proposed** networks included more regions of the anterior surface of the cornea. The anterior surface of the cornea has clear signs of KCN; therefore, **Composite** and **Proposed** could learn better features. In the bottom two OCT images of Fig. 7, the features activated by **Dense-L** had more background region below the posterior surface of the cornea. The background usually contains speckle noise, so **Dense-L** could be overfitting to data. On the other hand,

TABLE VIII
COMPARISON OF THE NUMBER OF LAYERS AND PARAMETERS OF DIFFERENT NETWORKS

Network	# Layers	# Parameters
Conv	22	14, 812, 995
Dense-S	23	48, 273, 219
Dense-M	24	50, 375, 491
Dense-L	24	86, 028, 099
Composite	23	48, 371, 523
Proposed	67	78, 587, 843

TABLE IX
COMPARISON OF THE TRAINING TIME PER EPOCH IN SECONDS, THE NUMBER OF TRAINING EPOCHS, AND THE TESTING TIME PER IMAGE IN SECONDS FOR DIFFERENT NETWORKS

Network	Training Time	# Training Epochs	Testing time
Conv	318.38	33.40	0.0090
Dense-S	333.83	37.80	0.0093
Dense-L	422.02	42.60	0.0094
Dense-L	459.46	45.20	0.0091
Composite	335.03	44.80	0.0098
Proposed	475.38	32.00	0.0147

the features activated by **Composite** and **Proposed** had the anterior and posterior surfaces of the cornea. These activated features were relevant to the cornea which ensured that **Composite** and **Proposed** learned better patterns associated with the corneal diseases. By comparing the saliency maps of **Composite** and **Proposed**, the features activated by **Composite** included some features related to the central artifact (e.g., The top first and third OCT images). This indicated that the **Proposed** network was less sensitive to noise. Using multiple image resolutions in the proposed network made the network more sensitive to the high-level features that were present at all resolutions and less sensitive to the lower-level features that were present only at the highest resolution.

Additionally, the results of the sensitivity analysis of the proposed network are shown in Fig. 7. The sensitivity heatmaps highlighted the anterior and posterior parts of the cornea for KCN, which is consistent with the changes caused by KCN to these parts. Also, the sensitivity heatmaps highlighted the posterior part of the cornea for FED, which is consistent with the changes caused by FED to the Descemet's membrane at the posterior part of the cornea. These visualizations confirm that the proposed network could learn feature patterns that were relevant to each disease.

D. Architecture and Time Comparison

We compared the architecture of all the networks based on the number of layers and parameters. The architecture comparison is summarized in Table VIII. The proposed network had less number of parameters compared to **Dense-L** because we used less fully connected layers with less neurons. If we sacrifice the gain in performance obtained from using multiple image resolutions, we could use **Composite** which outperformed **Dense-L**, but with significantly less number of parameters. The results of the training and testing times are summarized in Table IX. The proposed network converged faster to the final accuracy with

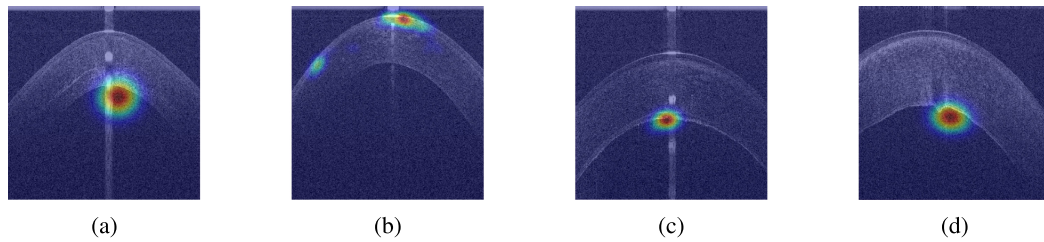


Fig. 8. The results of performing occlusion test on optical coherence tomography (OCT) images for the proposed network. The first two OCT images are from eyes diagnosed with keratoconus, and the last two images are from eyes diagnosed with Fuchs' endothelial dystrophy. The red and blue colors represent the most and least important regions, respectively. We used window sizes of (a) 101, (b) 25, (c) 75, and (d) 51 to obtain these visualizations.

the least number of epochs of 32 while **Dense-L** with the largest number of dense features converged with the largest number of epochs of 45. The proposed network had the largest mean running time per epoch, but its testing time was comparable to the other networks.

V. DISCUSSION

Optical Coherence Tomography (OCT) images are used in diagnosing different corneal diseases such as dry eye syndrome (DES), keratoconus (KCN), Fuchs' endothelial dystrophy (FED), and corneal graft rejection [17]–[20], [48]–[52]. In the literature, most of the deep learning methods used for corneal diagnosis focus on detecting KCN using corneal maps such as elevation, curvature, and pachymetry maps [39]–[41]. OCT images have only been used to directly detect graft detachment [42]. Signs of corneal diseases show in the OCT images; therefore, these diseases can be detected by training a network using OCT images. In [43] and [53], we fine-tuned pre-trained networks using OCT images to diagnose three corneal diseases: DES, KCN, and FED versus healthy controls (HCN). However, in this work, we focused on diagnosing two major corneal diseases, FED and KCN versus HCN. We proposed a novel network with a multi-resolution input, parallel resolution-specific encoders, and composite classification features to directly diagnose KCN and FED using OCT images. Using multiple input resolutions made the network process a larger receptive field of the input image. The encoders were implemented using the convolutional layers of a pre-trained VGG16 network. Resolution-specific feature maps were obtained by encoding the input at the different resolutions using parallel encoders. Multi-resolution feature maps were obtained by properly upsampling, concatenating, and reducing, depth-wise, the resolution-specific feature maps. Using parallel resolution-specific encoders helped to achieve multi-resolution feature fusion. The multi-resolution feature maps made the network more sensitive to the higher-level features and less sensitive to the lower-level noise (see Section IV-C). A fully connected layer was used to generate dense features. Composite classification features were obtained by concatenating the multi-resolution convolutional features and the dense features. Using composite classification features made the network learn better dense features, which were confirmed by the visualizations (see Section IV-A and Section IV-C). Using less dense features made the proposed network converge faster (see Section IV-D). We

solved the problem of class imbalance by oversampling the classes with fewer images. The oversampling was done by randomly sampling an equal number of images from each class in every batch of the data during the training (see Section II-A). For the experiments, we performed a k-fold cross-validation with $k=5$ to report the averaged performance of all the networks. To show the efficacy of the proposed network, we compared it with other networks that contained one or more of the components of the proposed network. We compared the networks by reporting the average of the ROC curves of each class and the mean $\pm$ standard deviation of the accuracy, sensitivity, specificity, F1-score, and AUC scores. The proposed network outperformed the other networks as reported in Section IV-A. To show the efficacy of the proposed network in classifying each disease, we reported the ROC curve for each class, computed the sensitivity, specificity, F1-score, and AUC scores, and computed the confusion matrix. The proposed network performed well for all classes with the best performance for FED and the least performance for HCN as reported in Section IV-B. The performance degradation in case of HCN is due to the fact that the OCT images of early-stage KCN have mild signs of the KCN disease and look similar to HCN. This was evident from the misclassification of HCN as KCN as reported in the confusion matrix in Section IV-B. To gain insight on how the networks work, we visualized them using saliency maps as described in II-D. The visualized heatmaps in Fig. 7 showed that the proposed network was sensitive to the high-level features of the OCT image that were related to the cornea. Also, we performed a sensitivity analysis of the proposed network to show the important regions in the input OCT image. The heatmaps in Fig. 7, obtained from the sensitivity analysis, showed that the anterior and posterior surfaces of the cornea were the most important regions in the OCT image, which is consistent with the changes that happen to the anterior and posterior surfaces due to the corneal diseases. To compare the architecture of the proposed network with the other networks, we reported the number of parameters, number of layers, and the training and testing times. Although the proposed network was the second-largest network in the number of parameters and the largest network in the number of layers, most of the parameters and layers of the proposed network were from the convolutional layers of the VGG16 network which were pre-trained. Therefore, our network took the least number of epochs to converge. The testing time of the proposed network was comparable to the other networks.

In this paper, we used only OCT images to predict corneal diseases and obtained promising results. Using OCT images along with other corneal maps such as thickness, elevation, curvature, and pachymetry maps can improve the results. This study is limited to the prediction of corneal diseases without determining the stage of each disease.

VI. CONCLUSION

In this paper, we proposed a novel network to directly diagnose two corneal diseases: Fuchs' endothelial dystrophy (FED) and keratoconus (KCN), using optical coherence tomography (OCT) images. The proposed network consists of a multi-resolution input, parallel resolution-specific encoders, and composite classification features. Using multiple input resolutions enlarges the receptive field processed by the network. We use the pre-trained convolutional layers of the VGG16 network as the encoder of our network. Resolution-specific feature maps are obtained by encoding the input resolutions using the parallel encoders. Multi-resolution feature maps are obtained by upsampling, concatenating, and reducing, depth-wise, the resolution-specific feature maps. The fused multi-resolution features made the network sensitive to the high-level features. Dense features are generated using a fully connected layer. Composite classification features are obtained by concatenating the convolutional and dense features. The composite classification features help to enhance the dense feature learning. A dataset of corneal OCT scans was obtained from patients recruited at the Bascom Palmer Eye Institute (BPEI). The OCT scans were diagnosed and labeled by corneal specialists at BPEI. The experiments showed that our proposed network outperformed other networks built with some components of our network. In addition, our proposed network was the fastest network to converge in training. The visualizations of the important features in some OCT images showed that the features were medically interpretable, and they highlighted the regions with clear signs of each disease. Finally, this study showed the efficacy of using deep learning for diagnosing corneal diseases directly from OCT images without the need for corneal maps such as thickness, elevation, and curvature maps.

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